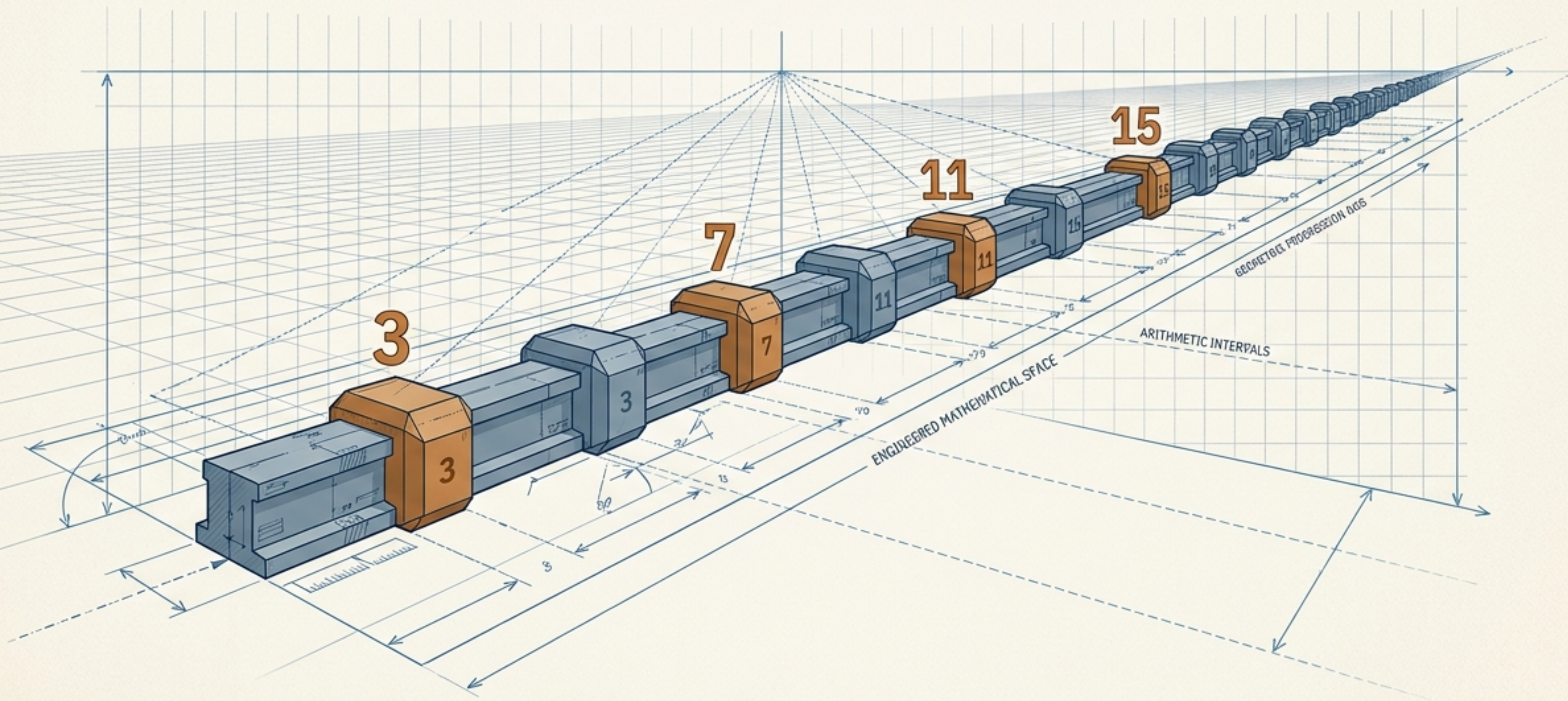


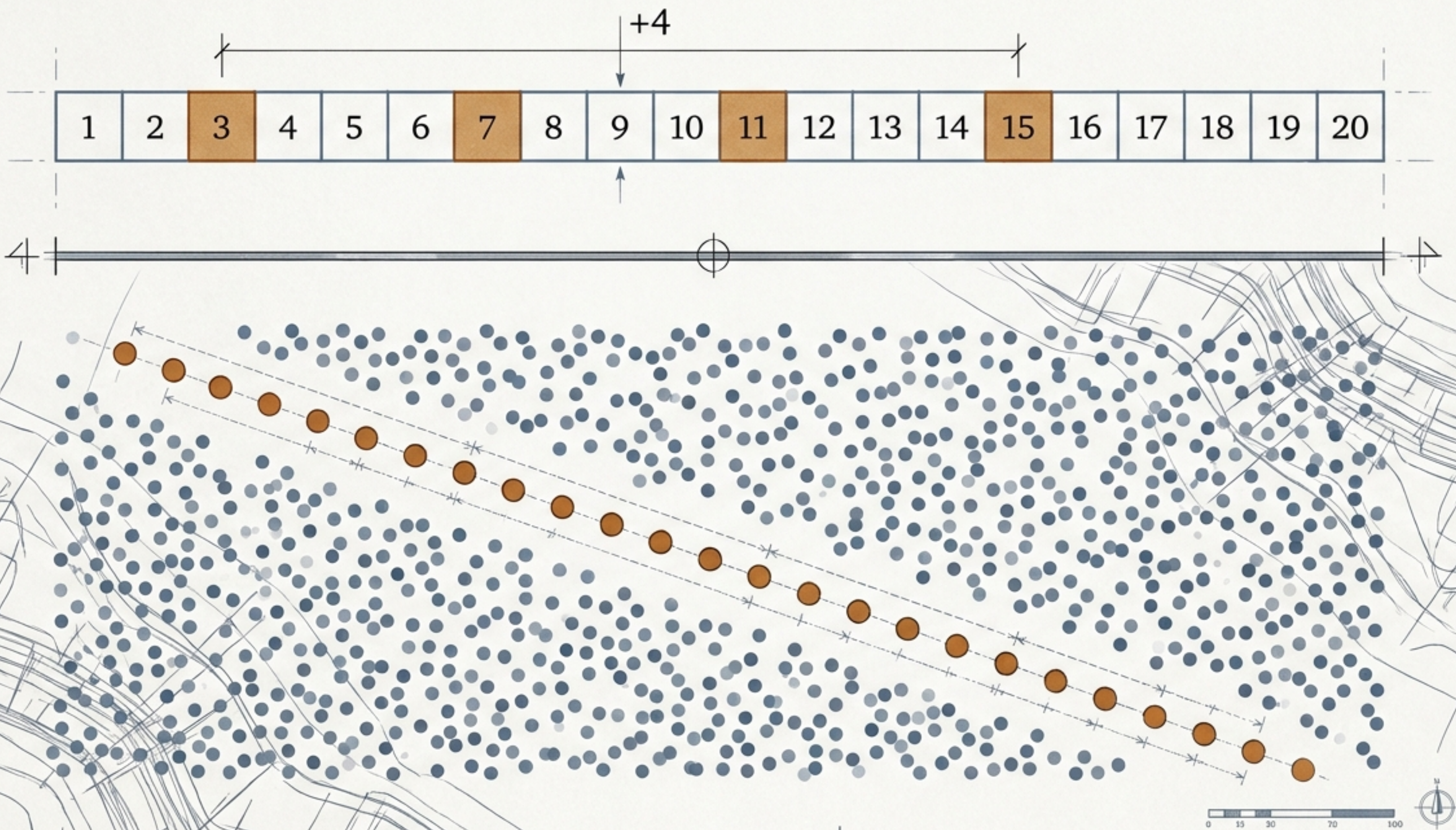
Breaking the Logarithmic Barrier

A visual guide to the OpenAI paper Quasipolynomial Bounds
for Arithmetic Progressions (September 2026)



This deck explains the AI-generated resolution to a nearly century-old problem in arithmetic combinatorics.

The search for infinite spacing



Definition

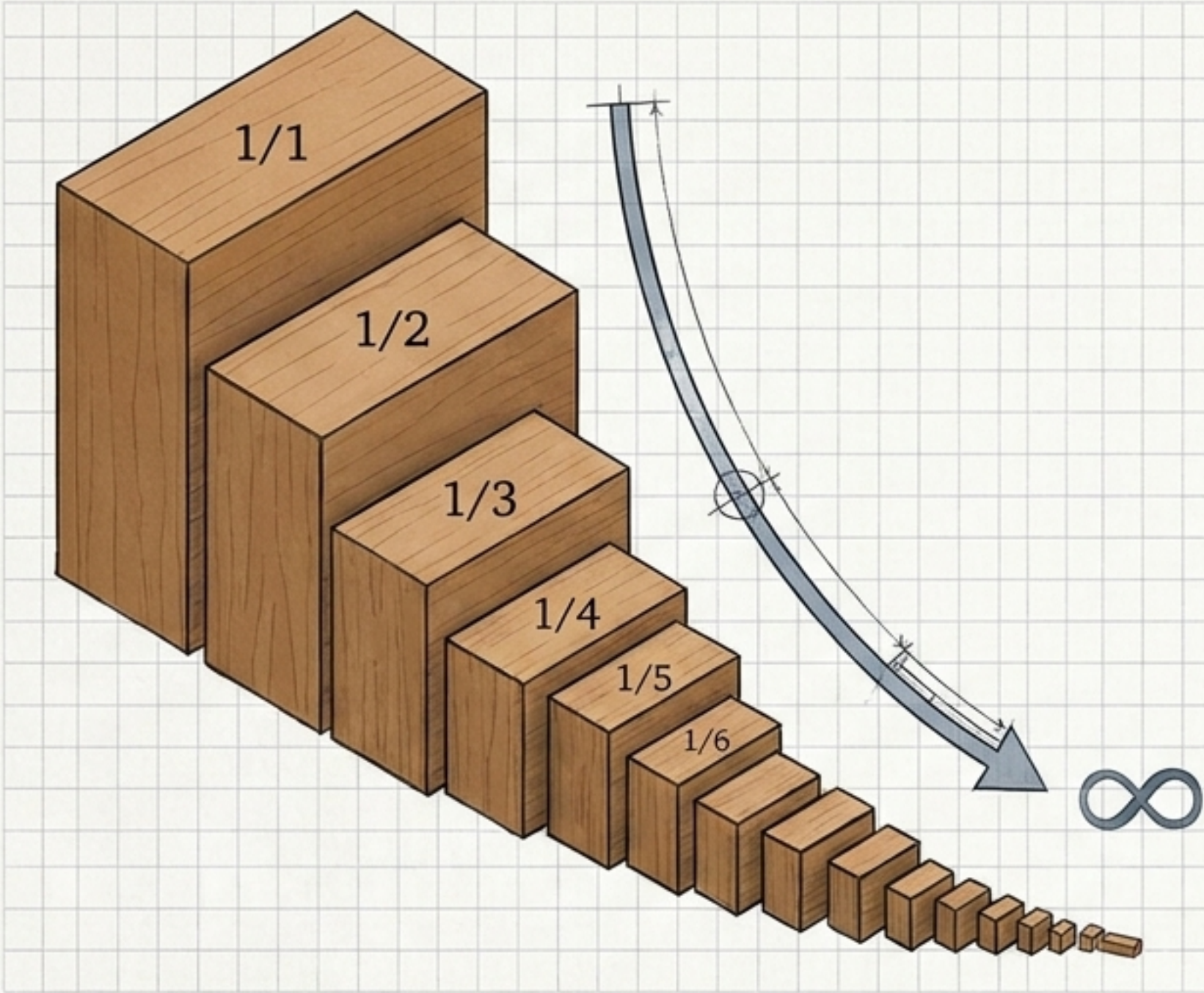
An Arithmetic Progression (AP) is a list of numbers separated by a constant, evenly spaced gap.

The Core Question

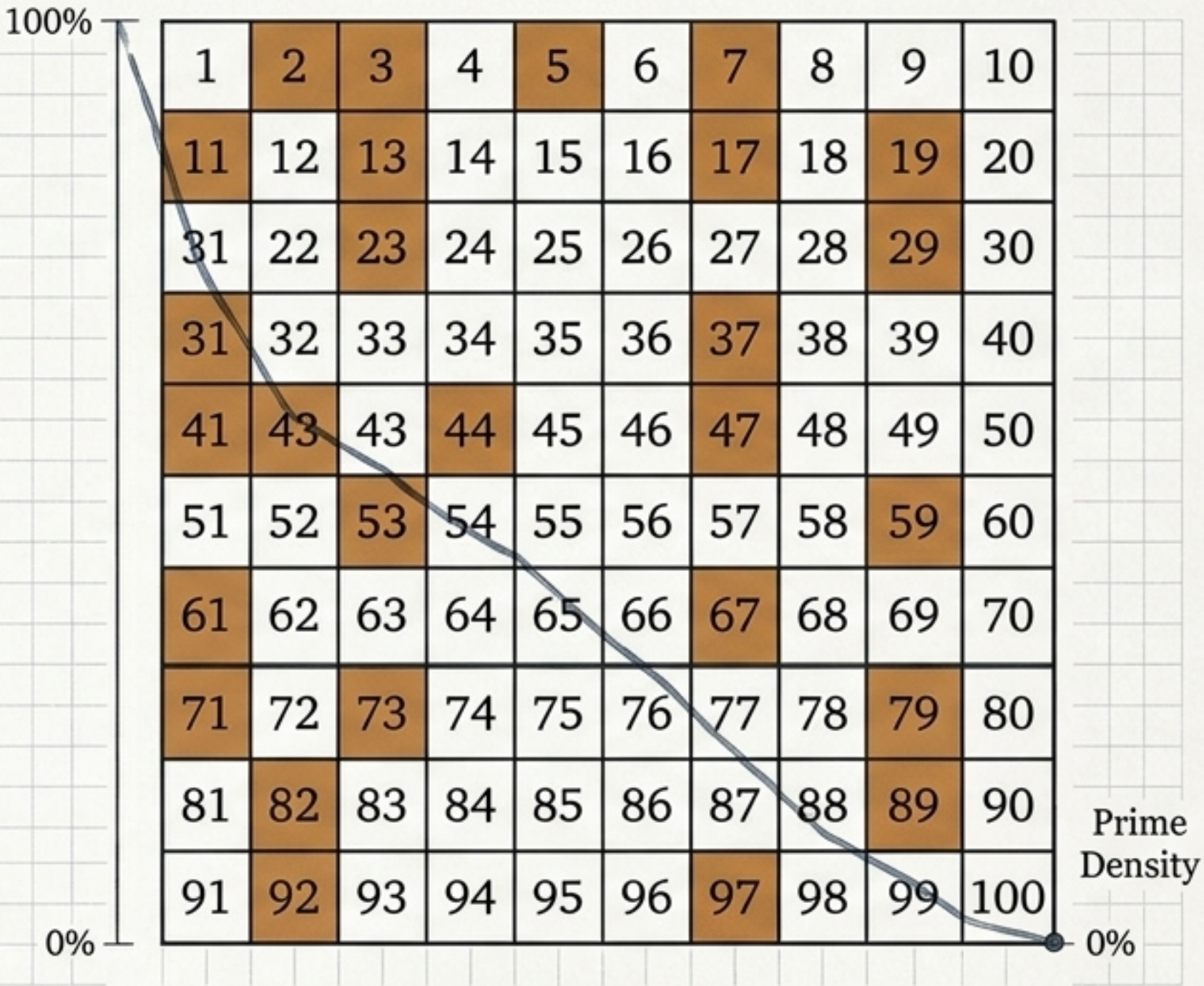
If you pick an arbitrary subset of integers, how large must it be to guarantee it contains an AP of any finite length k ?

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The Erdős test for mathematical largeness



In 1936, Paul Erdős posited a simple test: If the sum of the reciprocals of a set of positive integers diverges to infinity, the set contains APs of every finite length.

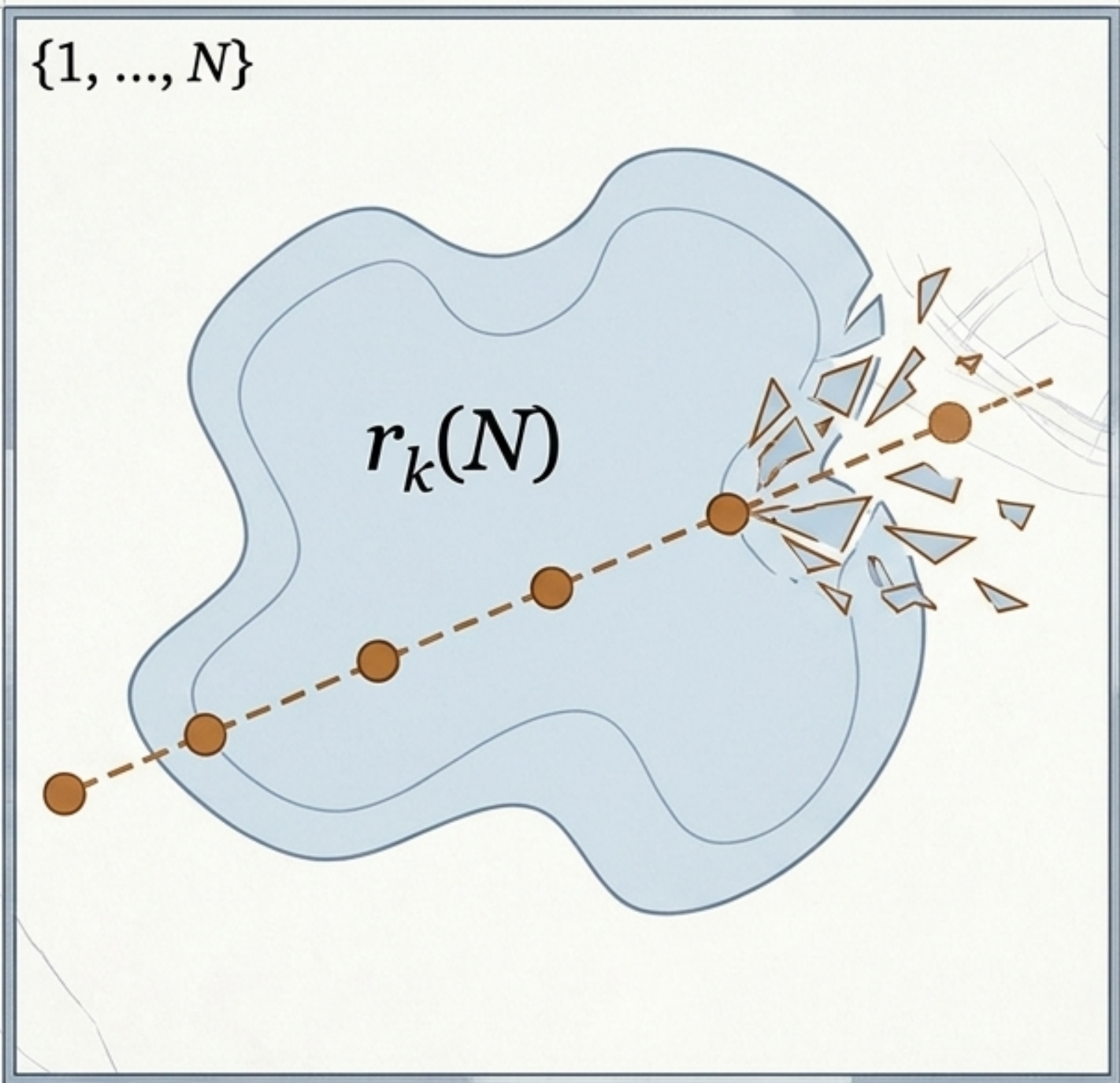


The ultimate test case was the prime numbers. Their density drops to zero, yet their reciprocals still sum to infinity. The general conjecture remained unsolved for decades.



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Measuring the avoidance of structure



To prove the

conjecture, mathematicians study sets that successfully avoid arithmetic progressions.

$r_k(N)$ = the size of the largest subset of numbers up to N with no k -term progression.

Szemerédi's Theorem (1975) proved the density of these avoiding sets drops to zero. But how fast does it drop?

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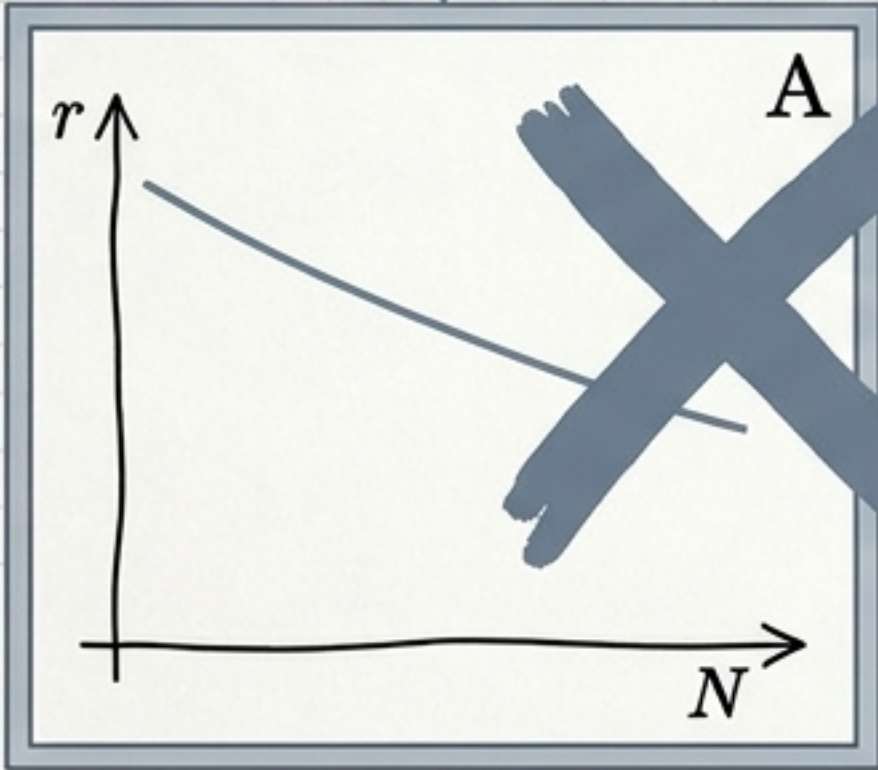
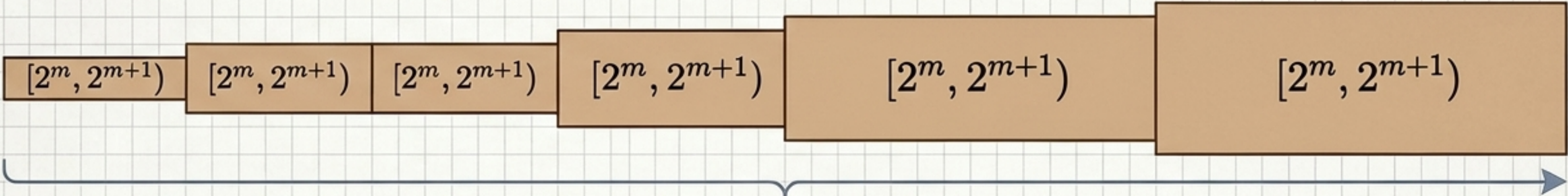
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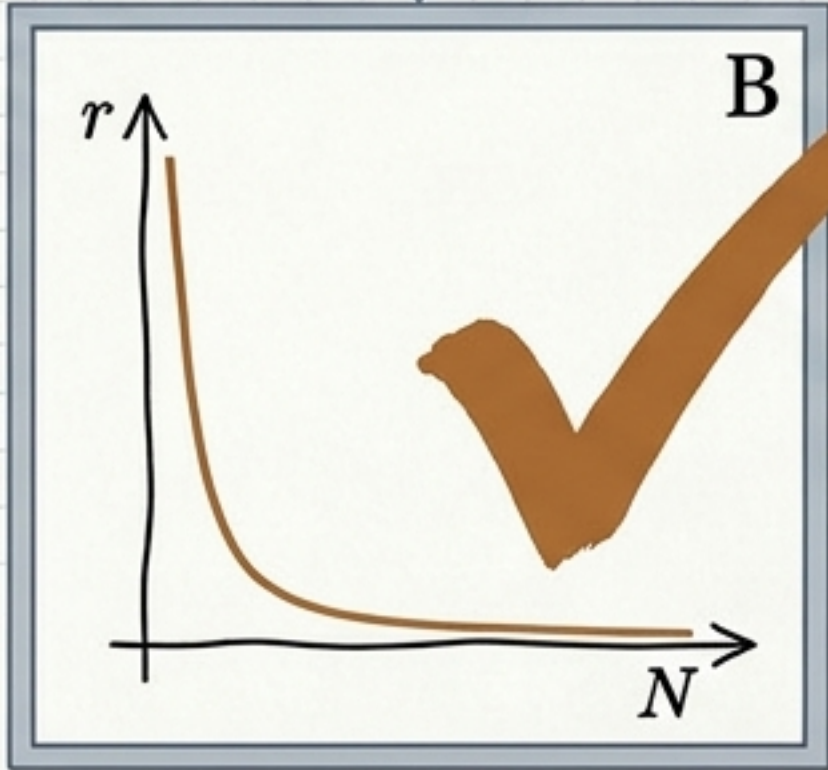
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The summability barrier requires a steep drop

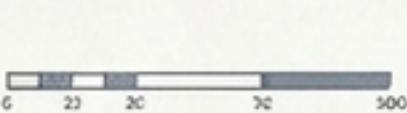
Reciprocal sum bound: $1 + \sum \frac{r_k(2^m)}{2^m}$



If the bound drops too slowly (e.g., $N / \log N$), the sum diverges.



We need a summable bound where the fraction shrinks fast enough that the sum converges.

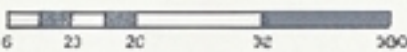


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A century-long relay race to break the barrier

Era	Mathematicians	Bound Formula	Summable?
1953	Roth	3-terms	No
2001	Gowers	$N/(\log \log N)^c$	No
2020	Bloom-Sisask	$N/(\log N)^{1+c}$	Yes (Settled $k=3$)
2023	Kelley-Meka	Stronger 3-term bounds	Yes
2024	Leng-Sah-Sawhney	$N \cdot \exp(-(\log \log N)^c)$ for $k \geq 5$	No
2026	OpenAI	?	?

Progress stalled at the logarithmic barrier. Finding a summable bound for all finite lengths remained out of reach.



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The quasipolynomial breakthrough

$$r_k(N) \leq C \cdot N \cdot \exp(-c(\log N)^\varepsilon)$$

Operational translation: A density of α forces a progression if $\log N \geq A(2 + \log 1/\alpha)^A$

In September 2026, an OpenAI preprint delivered a quasipolynomial threshold for every finite length $k \geq 3$.

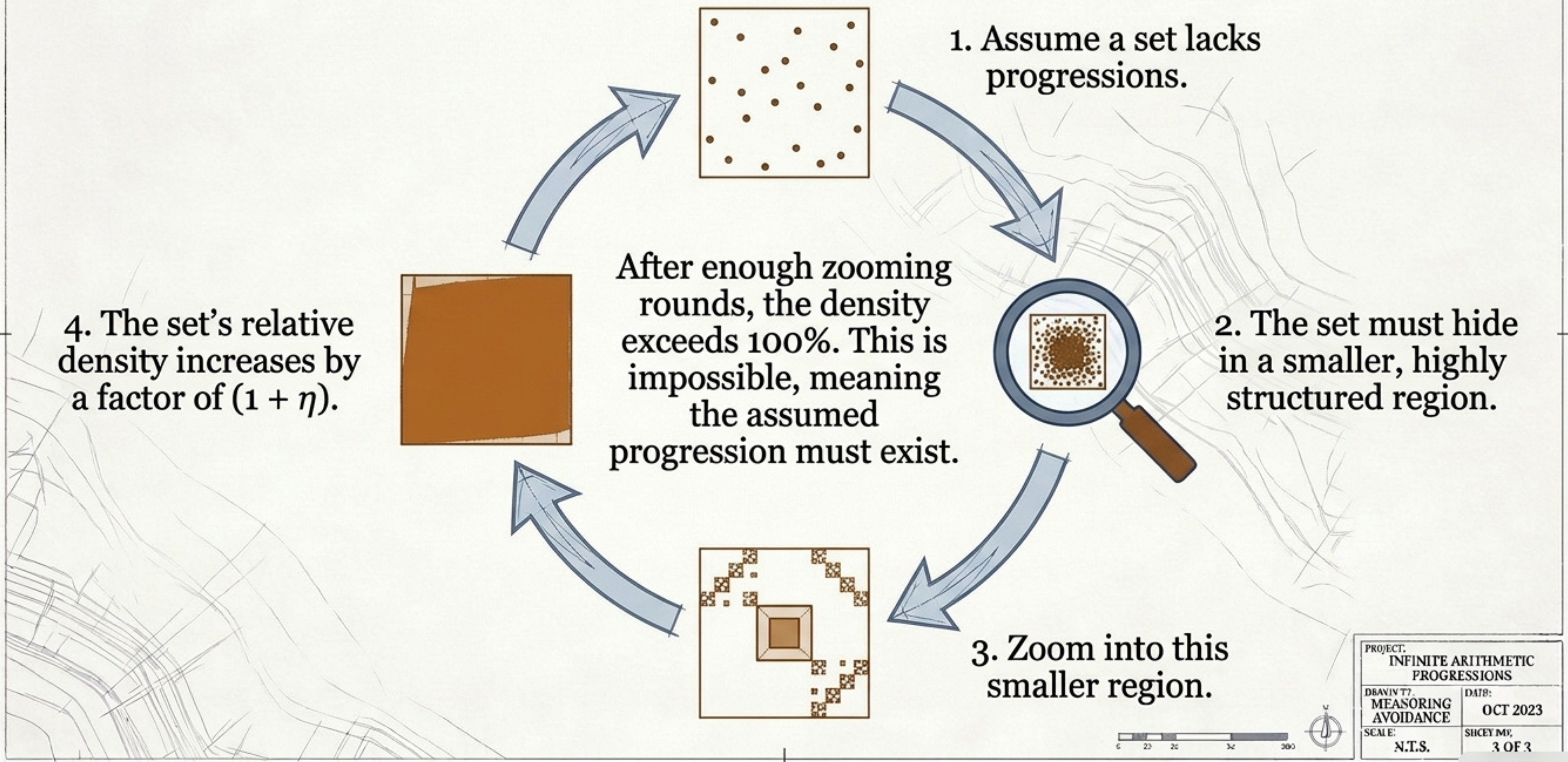
Crucially, this bound shrinks fast enough to be summable across dyadic blocks, officially proving the Erdős conjecture.

0 25 50 75 100



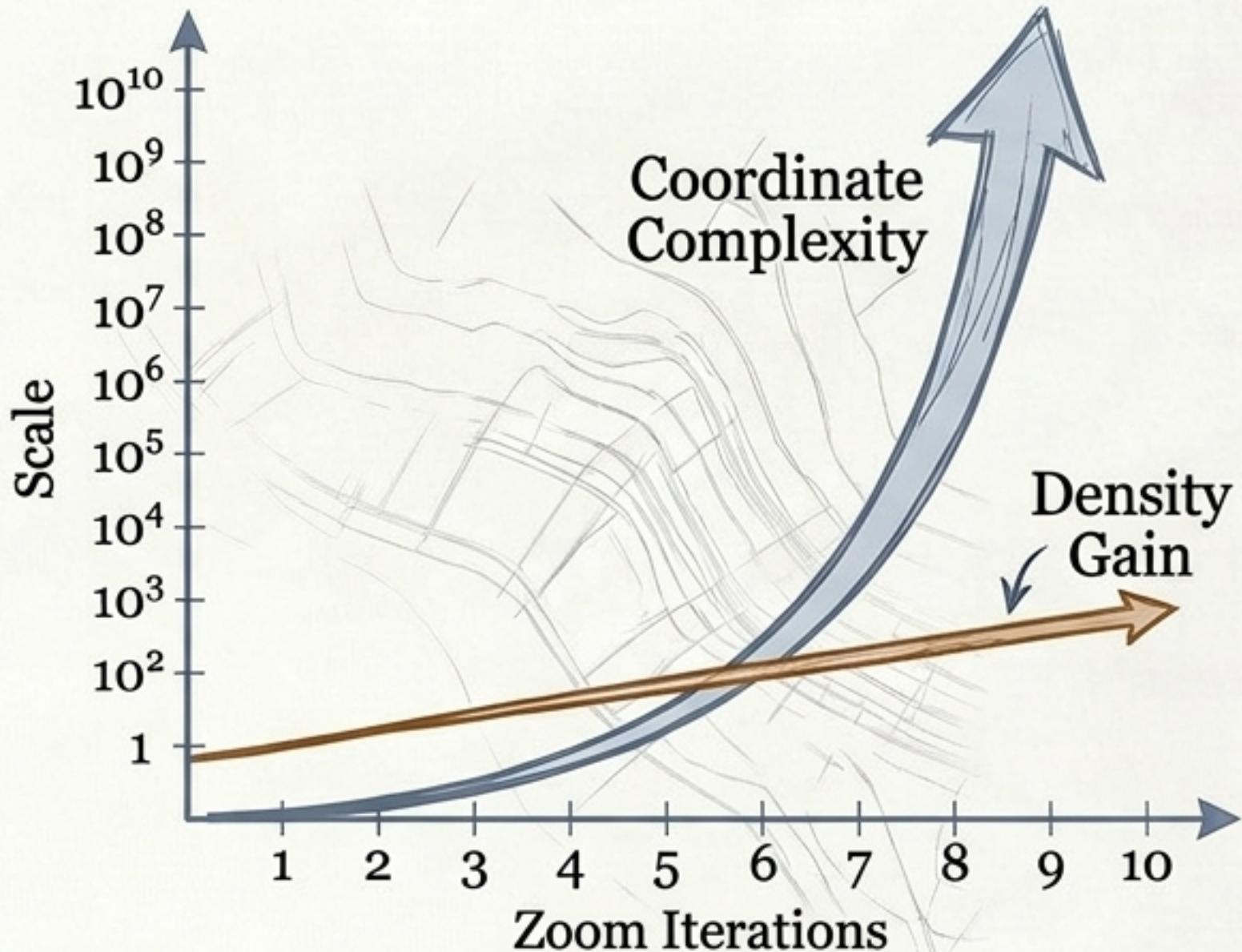
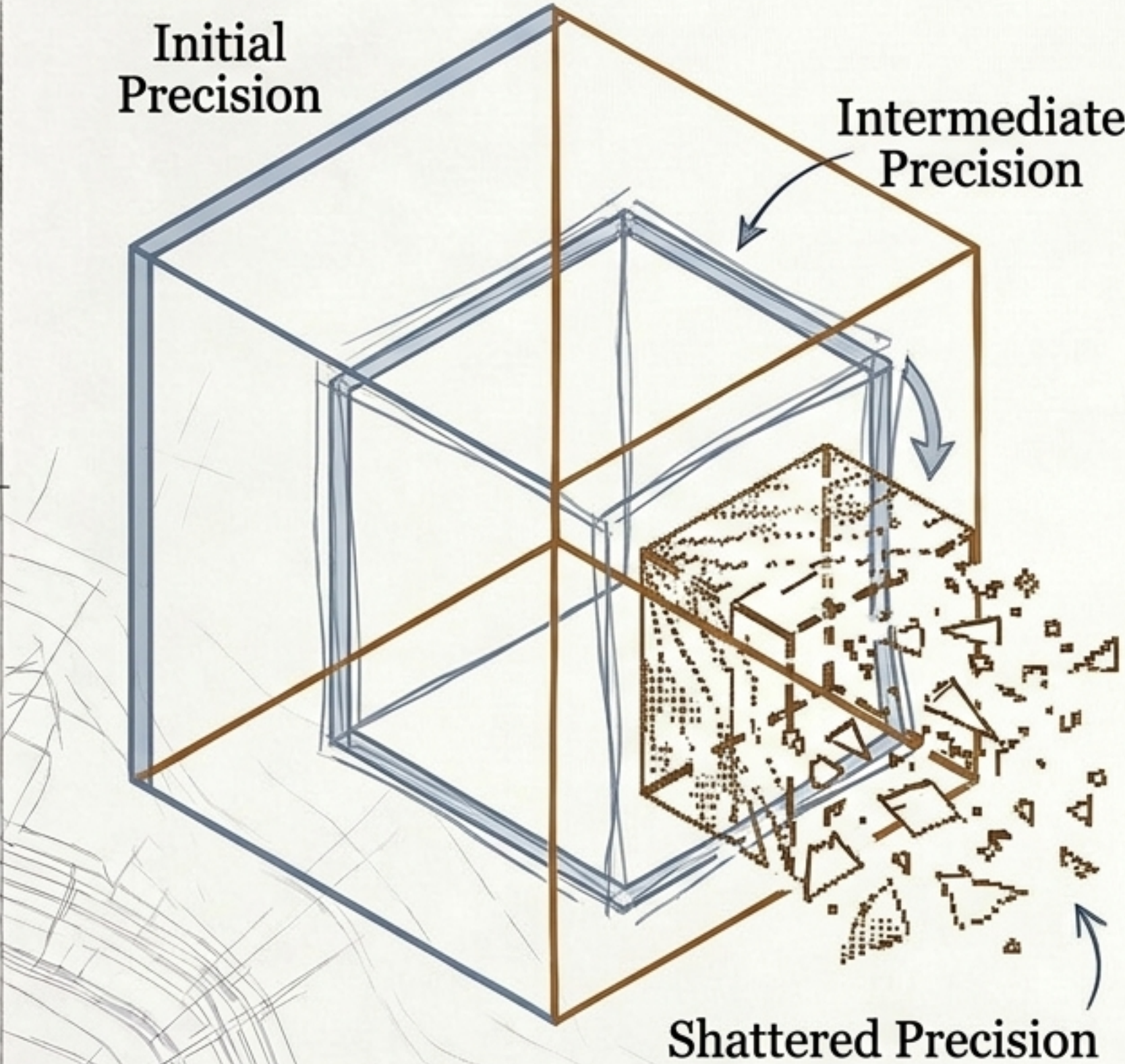
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The Contradiction Engine: Density-Increment Ladders



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The threat of exploding precision

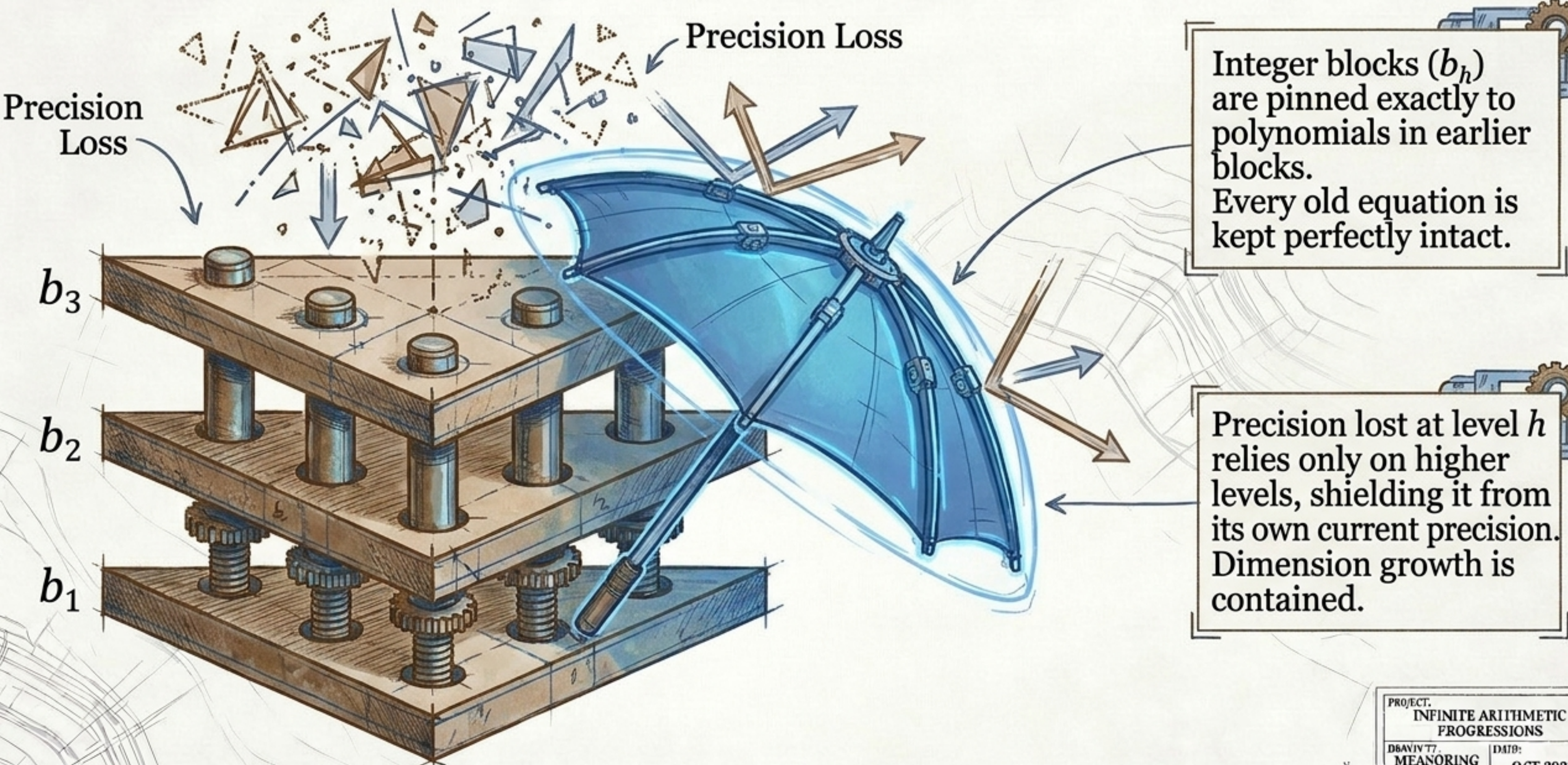


Every zoom requires complex coordinates to describe the new region. Historically, this precision cost compounds logarithmically, devouring the structural space before density reaches 100%.



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Triangular Polynomial Cells isolate precision loss



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Standing on the shoulders of giants

Leng–Sah–Sawhney

The inverse theorem for structured correlation.

**OpenAI
Increment
Engine**

Kelley–Meka

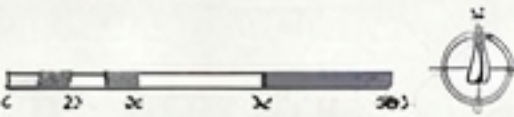
Shift comparison techniques (unbalancing and sifting).

Schoen–Sisask

Almost-periodicity to control loss in density increments.

The breakthrough requires the precise calibration of three recent masterworks to safely pass the density certificate down the stopped tree.

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Boundaries of the breakthrough

What it achieves

- ✓ Settles the Erdős reciprocal-sum conjecture for all finite lengths (k).
- ✓ Successfully solves the summability barrier with a quasipolynomial threshold.

What it doesn't do

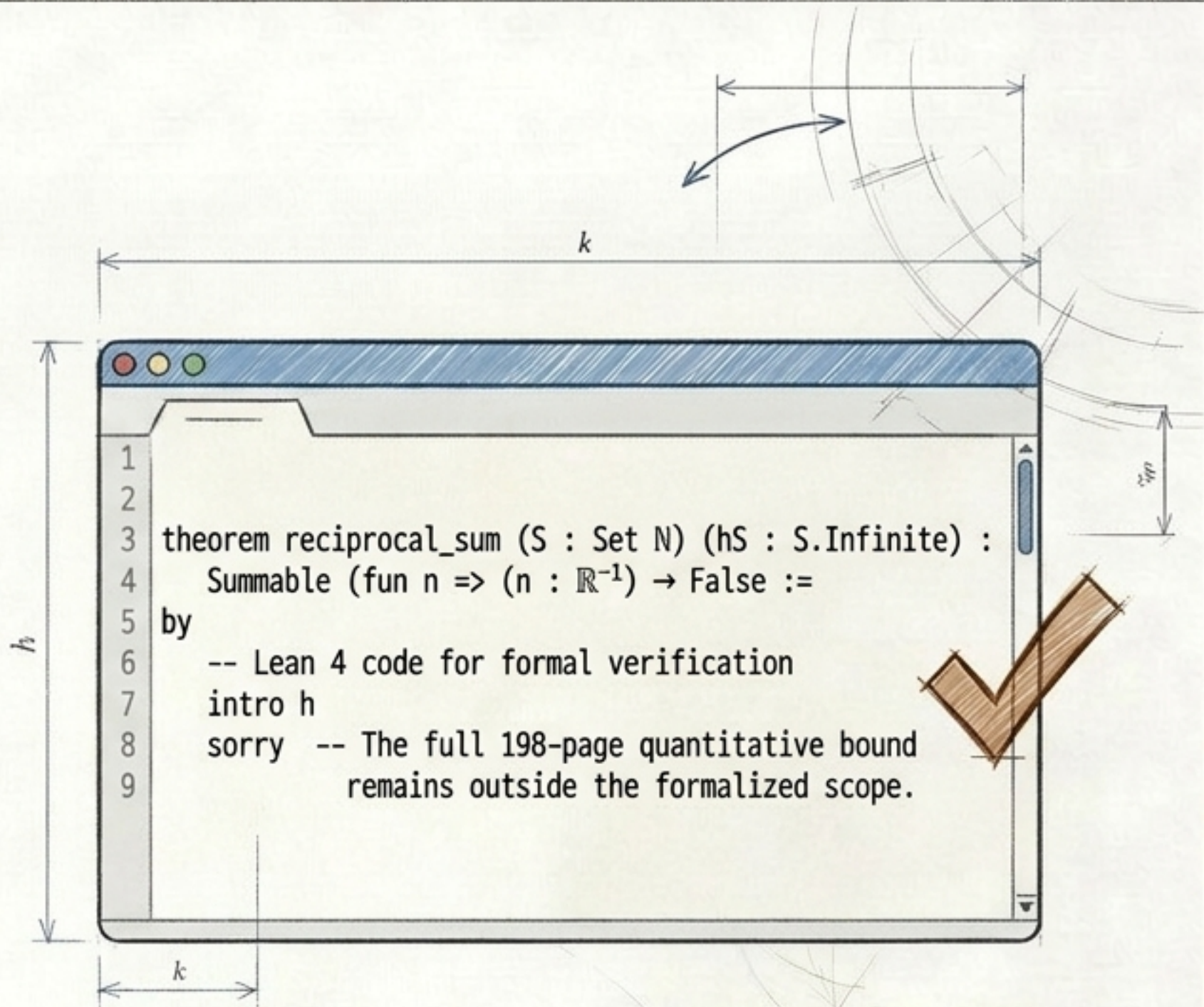
- Does not improve the specific 3-term bounds (Kelley-Meka's 2023 bound still holds).
- The constants (C, c, ε) are not mathematically explicit yet.

The paper optimizes for length (k) and establishing the ultimate summable bound, rather than tightening the explicit constants of the three-term case.

AI reasoning and formal verification



The Author: The 198-page preprint was generated by an OpenAI model, utilizing extensive internal mathematical reasoning trails.



The Verification: Lean formally covers the reciprocal-sum statement, but the full 198-page quantitative bound remains outside the formalized scope.

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The architecture of infinity mapped

$$\sum 1/n = \infty$$

By conquering precision loss in density increments, we have mapped the structure of integers deeper than ever before. If a set's reciprocals sum to infinity, it cannot hide. It must contain the universe of arithmetic progressions.

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